Requirements for the labs

Everything you need before Module 1, and what each module adds. Tick every box in sections 1 to 3 before the course starts; the per-module boxes can wait until the day before that module. The labs follow one company, Larkspur Mutual, from a single claims-intake agent to a platform of agents; one subscription and one tenant carry you through all seventeen modules.

1 · Accounts

- An **Azure subscription** where you can create resources and assign roles (Owner, or Contributor plus User Access Administrator, on a resource group). Learning path 1 runs on a free trial or a Visual Studio subscription; learning paths 2 to 4 need pay-as-you-go or an enterprise subscription, because Container Apps, API Management, Azure AI Search and Managed Redis stay up for days.
- Your **Microsoft account** that owns the Azure subscription. A personal account (outlook.com, hotmail.com) is fine for every module: sign in with it to the Azure portal, ai.azure.com and az login. A work or school account works the same way.
- A **GitHub account** for Module 9 (repository, GitHub Actions, environments) and Module 14 (evaluation gates). The free plan is enough.
- No **Microsoft 365 work account** is needed for the main course. Only Module 16 has an optional Teams step that needs one; its setup is in a separate guide, AI-500 - Teams setup for Module 16. Leave it until you want that step.

Which account you need, and where

What	Account	Used in	Required
Azure and Foundry	Your personal or work Microsoft account that owns the subscription	Every module	Yes
GitHub	A free GitHub account	Modules 9 and 14, only with APPLY=1	Optional; the labs print the commands instead
Teams and Power Automate	A Microsoft 365 work account with Teams and Power Automate	Module 16, Part 0 and Part 2, only with APPLY=1	Optional; see AI-500 - Teams setup for Module 16

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What	Account	Used in	Required
	Premium		
Entra ID P2 (Privileged Identity Management)	Not needed	Module 10 simulates just-in-time access locally	No
Microsoft 365 Copilot	Not needed	Not used in this course	No

NOTE If a Foundry or role-assignment step refuses your personal account, add a user to your subscription's directory (Azure portal, Microsoft Entra ID, Users, New user), give it Owner on rg-ai500, and sign in with that user from then on. It is free and needs no licence.

2 · Your computer

- Windows 11, macOS 13 or later, or Ubuntu 22.04 or later; 8 GB RAM; rights to install software.
- **Python 3.12 or later** (`python --version`). Every module runs in its own virtual environment.
- **Azure CLI 2.80 or later** (`az --version`). Sign in once with `az login`, then `az account set --subscription <id>`. All lab code is keyless: `DefaultAzureCredential` uses this session locally.
- **Visual Studio Code** with the Python extension and the Microsoft Foundry extension. **Git**, and the GitHub CLI (`gh`) for Module 9.
- **Docker Desktop** for Modules 6, 9 and 10 (MCP server containers, pipeline images). `az acr build` is the alternative when Docker cannot be installed.
- Network: outbound HTTPS to `*.azure.com`, `*.services.ai.azure.com`, `*.documents.azure.com` (Cosmos DB), `*.redis.azure.net`, `pypi.org` and `github.com`.

3 · Create once, use in every module

Do this the week before the course. Every module shares one resource group, one Foundry resource, one project and one Cosmos DB account.

1. Pick a region that offers every model and service the course uses: **East US 2** or **Sweden Central** (gpt-5, gpt-5-mini, text-embedding-3-large, Azure AI Search, Azure AI Content Safety, Cosmos DB serverless, Azure Managed Redis, Container Apps). Check the model availability table the week before; regions change.

2. Create the resource group and the Foundry resource, then the project: terminal az group create -n rg-ai500 -l eastus2 az cognitiveservices account create -n ai500-foundry -g rg-ai500 -l eastus2 \ --kind AIServices --sku S0 --custom-domain ai500-foundry --assign-identity # In https://ai.azure.com (new experience) open the resource, create the project "claims", # then copy the endpoint: https://ai500-foundry.services.ai.azure.com/api/projects/claims

```
terminal
az group create -n rg-ai500 -l eastus2
az cognitiveservices account create -n ai500-foundry -g rg-ai500 -l eastus2 \
  --kind AIServices --sku S0 --custom-domain ai500-foundry --assign-identity
# In https://ai.azure.com (new experience) open the resource, create the project
"claims",
# then copy the endpoint: https://ai500-
foundry.services.ai.azure.com/api/projects/claims
```

3. Deploy the models (Foundry portal, Build, Models, Deploy model; Global Standard): **gpt-5-mini** at 200K TPM or more, the default model for every agent; **gpt-5** at 50K TPM or more, used for critique and planning steps; **text-embedding-3-large** for Modules 7 and 8. Keep the deployment names equal to the model names; the `.env.example` files assume it.
4. Assign yourself the roles on the Foundry resource; they inherit to the project: terminal me=$(az ad signed-in-user show --query id -o tsv) scope=$(az cognitiveservices account show -n ai500-foundry -g rg-ai500 --query id -o tsv) az role assignment create --assignee $me --role "Foundry User" --scope $scope az role assignment create --assignee $me --role "Foundry Project Manager" --scope $scope az role assignment create --assignee $me --role "Cognitive Services User" --scope $scope # Module 11

```
terminal
me=$(az ad signed-in-user show --query id -o tsv)
scope=$(az cognitiveservices account show -n ai500-foundry -g rg-ai500 --query id -o
tsv)
az role assignment create --assignee $me --role "Foundry User" --scope $scope
az role assignment create --assignee $me --role "Foundry Project Manager" --scope
$scope
az role assignment create --assignee $me --role "Cognitive Services User" --scope
$scope # Module 11
```

5. Create the Cosmos DB account (serverless, NoSQL API) and give yourself the data-plane role. Used in Modules 1, 4, 8, 11 and 12: terminal az cosmosdb create -n ai500-cosmos -g rg-ai500 --locations regionName=eastus2 \ --capabilities EnableServerless az cosmosdb sql role assignment create -a ai500-cosmos -g rg-ai500 --principal-id $me \ --role-definition-id 00000000-0000-0000-0000-000000000002 \ --scope $(az cosmosdb show -n ai500-cosmos -g rg-ai500 --query id -o tsv) # 0000...0002 is "Cosmos DB Built-in Data Contributor"; the endpoint is https://ai500-cosmos.documents.azure.com

```
terminal
az cosmosdb create -n ai500-cosmos -g rg-ai500 --locations regionName=eastus2 \
  --capabilities EnableServerless
az cosmosdb sql role assignment create -a ai500-cosmos -g rg-ai500 --principal-id $me \
  --role-definition-id 00000000-0000-0000-0000-000000000002 \
  --scope $(az cosmosdb show -n ai500-cosmos -g rg-ai500 --query id -o tsv)
# 0000…0002 is "Cosmos DB Built-in Data Contributor"; the endpoint is https://ai500-
cosmos.documents.azure.com
```

6. Connect Application Insights: Foundry portal, Observability, Tracing, Connect, creating a new resource in rg-ai500. Used from Module 1 (run traces) and in every module of learning path 4.

CHECK After the Module 1 setup, python section-1/first_loop.py prints one completed run with its status transitions and writes a run document to Cosmos DB. If it does, the shared setup is complete.

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4 · What each module adds

Each module's lab guide repeats its own line. Resources marked shared are the ones from section 3.

MODULE	RESOURCES	ROLES AND QUOTA
LEARNING PATH 1 · ARCHITECT		
1 · Stateful agentic loops	Cosmos DB (shared) database larkspur, containers runs and checkpoints (partition key /sessionId, TTL on); Application Insights (shared)	Cosmos DB Built-in Data Contributor; gpt-5 for the critique step
2 · Orchestration patterns	Nothing new	gpt-5-mini at 200K TPM or more: five specialists run in parallel
3 · Task decomposition	Nothing new	—
4 · Agent communication with A2A	Azure Managed Redis (Balanced B0); Cosmos DB container registry	Redis access policy Data Contributor for your account (Entra authentication)
LEARNING PATH 2 · BUILD		
5 · Advanced prompting	Azure AI Content Safety (Prompt Shields) through the Foundry resource; Foundry evaluations	Cognitive Services User
6 · Tool ecosystems with MCP	Container Apps environment, Container Registry (Basic), API Center (Free)	AcrPush; a user-assigned managed identity per MCP server
7 · Advanced RAG	Azure AI Search (Basic, semantic ranker on); text-embedding-3-large (shared)	Search Service Contributor and Search Index Data Contributor
8 · Memory architectures	Cosmos DB (shared) with vector indexing enabled; container memory partitioned by policyholder	—
LEARNING PATH 3 · DEPLOY AND GOVERN		

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MODULE	RESOURCES	ROLES AND QUOTA
9 · CI/CD with GitHub Actions	GitHub repository with dev, staging and prod environments; Container Registry and Container Apps (from Module 6)	Federated (OIDC) identity for Actions with Contributor on rg-ai500
10 · Zero-trust security	Key Vault, one user-assigned managed identity per agent, private endpoint for Cosmos DB; AKS Automatic optional	Key Vault Secrets Officer; Owner on rg-ai500 to assign per-agent roles
11 · Responsible AI governance	Content Safety (shared); Storage account for protected trace export; Azure Policy assignment	Storage Blob Data Contributor
12 · Agent lifecycle governance	API Management (Consumption) in front of the model deployments; Cost Management export	Cost Management Reader on the subscription
LEARNING PATH 4 · MONITOR, EVALUATE AND OPERATE		
13 · Distributed observability	Application Insights and Log Analytics (shared); one workbook	Monitoring Contributor
14 · Evaluation frameworks	Foundry evaluations; the GitHub repository from Module 9	gpt-5 as the judge model
15 · Performance and cost	Managed Redis (from Module 4) as the semantic cache; gpt-5-nano deployment	—
16 · Human-in-the-loop approvals	Power Automate flow and a Teams channel in your Microsoft 365 tenant	Microsoft 365 work account; optional
17 · Production incidents	Storage account (replay snapshots, from Module 11); Azure Monitor alert rules and action group; one Logic App runbook	—

5 · Cost and clean-up

Running every lab once costs a few tens of dollars at pay-as-you-go prices. The largest items are gpt-5 tokens, one Azure AI Search Basic service (about 75 USD per month while it exists), Azure Managed Redis B0, API Management Consumption and Container Apps (which scale to zero). Cosmos DB serverless and Application Insights cost cents at lab volumes. Delete everything at the end of the course:

```
terminal
az group delete -n rg-ai500 --yes --no-wait
```

NOTE Instructors: complete sections 1 to 3 and the Module 9 and 10 role assignments for the class before day one. Those steps need Owner rights that students often do not have.

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